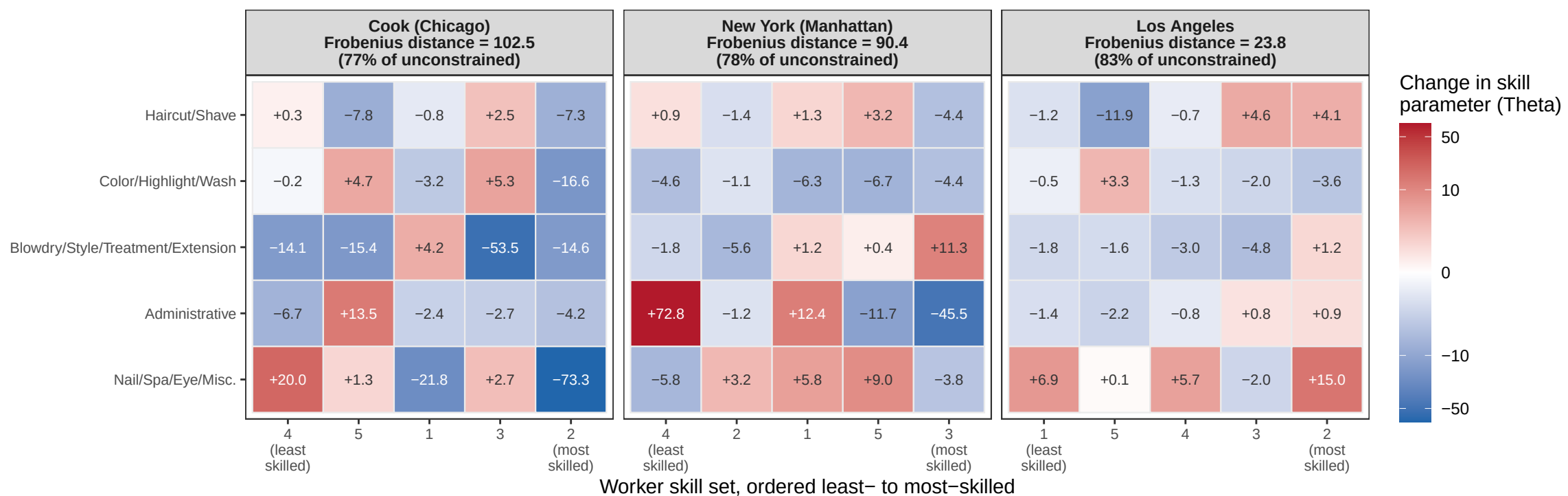




Note: Each cell is the change in the estimated skill parameter Theta from imposing the constraint (constrained minus unconstrained) for one worker skill set (column) at one task (row): red means the constraint raised the estimate, blue that it lowered it. Unconstrained estimates are the baseline; constrained estimates add the requirement that skill sets can be ranked by absolute advantage, i.e. that one ordering exists in which each skill set is at least as skilled as the previous one at every task. Columns are ordered least- to most-skilled under that ranking, so under the constraint values never fall from left to right; skill sets are numbered as in the paper and the ordering differs by county. The colour scale is compressed logarithmically (preserving sign) and shared across counties. Each heading reports the Frobenius distance between that county's unconstrained and constrained matrices (the square root of the summed squared cell-by-cell changes), with that distance as a percentage of the unconstrained matrix's own Frobenius norm.